

Abdoul Hamid COULIBALY

Backend Engineer — Data Systems & Cloud

📍 Rennes, France

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Software engineer working at the intersection of data, cloud infrastructure, and distributed systems. After several years building large-scale data processing architectures (IoT, telecom, renewable energy), I now focus on Infrastructure as Code components around Terraform, with a growing contribution to open source projects.

Professional Experience

Software Engineer – Open Source Terraform Provider

June 2026 - Present

INETUM (Orange Business) | Rennes, France

Cloud Avenue Terraform Provider

- Took over technical ownership of the Cloud Avenue Terraform provider, published on the Terraform Registry for platform customers.
- Ongoing development, maintenance, and improvement: Terraform resources and data sources, Go SDK, releases, and documentation.
- GitHub triage and support: user issues, feature requests, and guidance for external contributors.

Open Source Terraform Plugins

- Maintenance and evolution of 4 cross-provider Terraform plugins in the GitHub organization, used by external organizations including Microsoft.
- Open source community support: code reviews, issue triage, and contributor guidance.

Go Terraform HCL OpenAPI AI Agentic Development Git GitHub Actions CI/CD Open Source

Backend Engineer – Data Supply Chain

November 2023 - March 2026

Univers (QoS Energy) | La Chapelle-sur-Erdre, France

Architecture & Data Ingestion

- Development and refactoring of Go and Python microservices for ingesting millions of metrics per minute, in a GitOps-managed Kubernetes environment.
- Implementation and extension of event-driven architectures (RabbitMQ, NATS) for ingestion and asynchronous processing.
- Design of resilient pipelines handling network issues and source instability: distributed error handling, retry, backfilling and manual reload.

Integration & Connectors

- Development of connectors for heterogeneous sources: SCADA, REST, FTP, SQL databases.
- Active participation in architecture decisions: technology evaluation (NATS, Redis), data modeling and source API rate limiting management.
- Rewriting Go services to Python to align with Bazefield stack constraints.

Scalability & Resilience

- Service instrumentation with Prometheus and Grafana dashboard setup for operational monitoring.
- On-call rotation (1 week/6): critical incident resolution and L3 support for client technical operations.

Go Python Ruby RabbitMQ NATS PostgreSQL Kubernetes Azure GitOps Prometheus Grafana



Software Engineer

NumoData (formerly EXFO Solutions) | Rennes, France

Backend Development

- Development of Go backend services for a network monitoring platform processing millions of metrics per minute.
- Design of analytical queries (SQL, Druid): aggregations, multidimensional queries, optimization via indexing and caching.

Smart Usage Monitoring — Technical Ownership

- End-to-end design and development of a dashboard usage analytics service: most viewed metrics, highest-volume dimensions, detection of underutilized data.
- Results exposed via REST API, Prometheus instrumentation and Grafana dashboards, used for cache optimization and storage cost reduction.

Performance & Infrastructure

- Load testing and correlation analysis of volume/resources (CPU, RAM, latency) for client sizing.
- Development of an internal tool for automatic Kubernetes (Helm) configuration generation based on client sizing results.

Go

Python

Kafka

Apache Druid

AWS

Prometheus

Grafana

Helm

Kubernetes

October 2021 - October 2023



Web Developer – Internship

Resalocal | Talloires-Montmin, France

- Development of an ingestion pipeline from the Apidae tourism database to the Akeneo PIM: import of POIs, events and activities.
- Bidirectional synchronization: returning enriched data to Apidae after processing.
- POI media management via Cloudinary API: image upload and transformation.
- Handling ingestion constraints: pagination, rate limiting, and preservation of existing PIM data.

PHP

Laravel

Docker

Kubernetes

Akeneo

Cloudinary

REST API

May 2021 - August 2021

Technical Skills



Languages

Go

Python

Ruby

TypeScript

SQL

Bash



Cloud & DevOps

AWS

Azure

Docker

Kubernetes

Terraform

Helm

GitLab CI



Data & Messaging

PostgreSQL

MySQL

Apache Druid

Kafka

RabbitMQ

NATS



Observability

Prometheus

Grafana

OpenTelemetry

Education

Master's in Computer Science – DABI (Decision-Making, Learning & Big Data)

University of Rennes 1

Data warehousing, data mining, Cloud & Big Data, artificial intelligence, application integration

2021 – 2023

Bachelor's in Computer Science – IT Management

University of Rennes 1

IT/management dual competency, databases, web development, object modeling, data analysis

2020 – 2021

Higher Technician Diploma in Computer Science

INP-HB

Algorithms, networking, web development, system administration, project management

2016 – 2019

General & Professional English Training

Wall Street English

Oral and written communication, IT technical vocabulary, professional presentations, meetings in English

2024

Languages

French (native)

English (fluent)

Bambara (native)